The Laryngoscope

Recurrent Laryngeal Nerve Regeneration Using an Oriented Collagen Scaffold Containing Schwann Cells

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Objectives/Hypothesis: Regeneration of the recurrent laryngeal nerve (RLN), which innervates the intrinsic laryngeal muscles such that they can perform complex functions, is particularly difficult to achieve. Synkinesis after RLN neogenesis leads to uncoordinated movement of laryngeal muscles. Recently, some basic research studies have used cultured Schwann cells (SCs) to repair peripheral nerve injuries. This study aimed to regenerate the RLN using an oriented collagen scaffold containing cultured SCs.

Study Design: Preliminary animal experiment.

Methods: A 10-mm-long autologous canine cervical ansa was harvested. The nerve tissue was scattered and subcultured on oriented collagen sheets in reduced serum medium. After verifying that the smaller cultivated cells with high nucleus-cytoplasm ratios were SCs, collagen sheets with longitudinally oriented cells were rolled and inserted into a 20-mm collagen conduit. The fabricated scaffolds containing SCs were autotransplanted to a 20-mm deficient RLN, and vocal fold movements and histological characteristics were observed.

Results: Scaffolds containing cultured SCs were successfully fabricated. Immunocytochemical examination revealed that these isolated and cultured cells, identified as SCs, expressed S-100 protein and GFAP but not vimentin. The orientation of SCs matched that of the oriented collagen sheet. Two months after successful transplantation, laryngeal endoscopy revealed coordinated movement of the bilateral vocal folds by external stimulation under light general anesthesia. Hematoxylin and eosin staining showed that the regenerated RLN lacked epineurium surrounding the nerve fibers and was interspersed with collagen fibers. Myelin protein zero was expressed around many axons.

Conclusions: Partial regeneration of RLN was achieved through the use of oriented collagen scaffolding.

Key Words: Recurrent laryngeal nerve, Schwann cells, regeneration, collagen tube, cell orientation.

Level of Evidence: NA

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INTRODUCTION

Peripheral branches of cranial nerves differ substantially from the central nervous system in their ability to regenerate after injury. Repair of peripheral nerves following an injury typically occurs naturally in short segments. However, if the damage is too severe for regeneration to take place, surgical interventions such as end-to-end suturing or autografting[1] of the peripheral nerve may be required. Unfortunately, reconstructive surgery for the recurrent laryngeal nerve (RLN), which innervates the intrinsic laryngeal muscles such that they can perform complex functions, can yield incomplete or ineffective functional regeneration and can cause autografting complications such as donor site morbidity.[2–5] Synkinesis after RLN neogenesis can also result in uncoordinated movement of laryngeal muscles. Recently, various types of nerve guidance conduits (NGCs) have been found to improve axonal regeneration from the proximal stump to the distal stump.[6–11] The NGCs can create a favorable microenvironment spanning a greater distance of reinnervation.

Schwann cells (SCs) are required to repair peripheral nerves following injury, and SC proliferation can form cellular columns known as bands of Büngner,[7] which act as guidance pathways for axonal regeneration. The cones of the axon from the proximal stump then extend through the pathway at a rate of up to 1 mm per day.[8] In the past decade, some basic research studies have used cultured SCs to repair peripheral nerve injuries. However, transplantation of NGCs including cultured SCs that have grown in multiple directions provides no spatial guidance to improve axonal alignment. Successful remyelination of established axons following injury may require cultured SCs that are oriented longitudinally from the proximal to distal axons early on, before degeneration of the distal axons occurs.

We hypothesize that culturing SCs with cells that are oriented longitudinally from the proximal to distal axons will facilitate axonal regrowth with the proper

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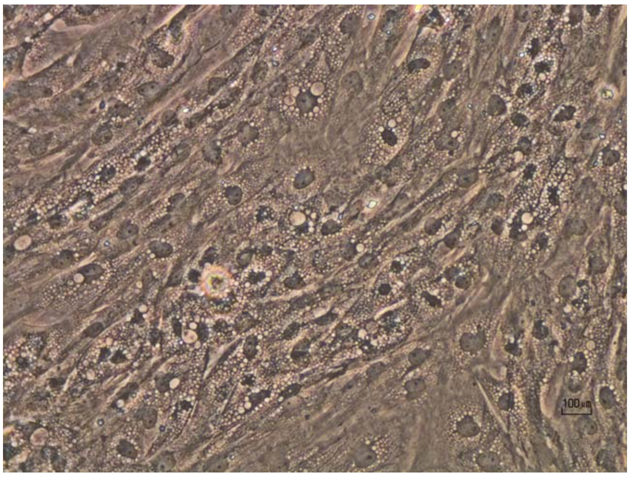

Fig. 1. Phase-contrast microscopy of the cultivated cells. Smaller cells with high nucleus-cytoplasm ratios grew out from the peripheral nerve fragment after a week of primary culture.

alignment of RLN fibers. The purpose of this study was to regenerate functional RLN using an oriented collagen scaffold containing cultured SCs.

MATERIALS AND METHODS

All experimental protocols used in this study were approved by the Kurume University Animal Care and Treatment Committee.

Cell Harvesting, Culture, and Isolation

Three experimental beagle dogs were employed in the current study. They were anesthetized by intramuscular administration of xylazine (2 mg/kg) and midazolam (0.3 mg/kg), and intravenous administration of pentobarbital sodium (5–10 mg/kg). A 10-mm-long autologous cervical ansa was surgically excised from each canine. The specimens, excluding the epineurium, were washed with Dulbecco's phosphate-buffered saline (PBS) containing 1% penicillin/streptomycin (pen/strep), and incubated in Dulbecco's Modified Eagle's Medium/Nutrient Mixture F-12 (DMEM/F12) (GIBCO, Waltham, MA) containing 1.2 U/mL of dispase (Roche, Basel, Switzerland) and 0.5 mg/mL of collagenase (Sigma-Aldrich, St. Louis, MO) at 37°C for 1 hour. The supernatant was removed after centrifugation at 400*g* for 5 minutes. Specimens were incubated in additional DMEM/F-12 and 10% fetal bovine serum (FBS) (GIBCO) containing 1% pen/strep at 37°C for 5 minutes. Each collected specimen was divided into sections of 1 to 2 mm using a sterile scalpel under a stereoscopic microscope. These sections were placed on a 100-mm dish and cultured in DMEM/F-12 and 10% FBS containing 1% pen/strep at 37°C in a humidified atmosphere of 5% CO_2. Fibroblast growth could be identified first under a phase-contrast microscope (Nikon, Tokyo, Japan) within a week of primary culture. Thereafter, smaller cultivated cells with high nucleus-cytoplasm ratios could be identified contiguously with fibroblasts (Fig. 1). At that point, DMEM/F-12 and a low-serum medium of 2.5% FBS containing 1% pen/strep was provided for these smaller cells for the next few weeks. These cultivated cells with semiconfluent growth were isolated as single-cell suspensions using a cell scraper.

Immunocytochemical Investigations of the Isolated Cells

Isolated cells grown on the chamber slides were fixed by 4% paraformaldehyde in 0.1 M PBS for 15 minutes, after which the medium was aspirated and cells were rinsed with PBS. Specimens were pretreated in 10 mM glycine in PBS and 3% bovine serum albumin to block residual reactive aldehydes and nonspecific interactions, respectively. The specimens were then incubated individually at 37°C for 40 minutes with the following primary antibodies: anti-S100b (ab11179; Abcam, Cambridge, UK) diluted 1:500, anti-glial fibrillary acidic protein (GFAP) (ab10062; Abcam) diluted 1:100, and anti-vimentin (ab8978; Abcam) diluted 1:500. After rinsing with PBS and incubating with secondary antibodies conjugated with Alexa Fluor 674 (Life Technologies, Darmstadt, Germany) diluted 1:1,000 at 37°C for 40 minutes, the immunoreactive proteins were stained with 4′,6-diamidino-2-phenylindole (Vector Laboratories, Burlingame, CA). Immunoreactivity was examined by fluorescence microscopy (Nikon).

Fabrication of the Oriented Collagen Scaffold With Cultured Schwann Cells

After verifying that the obtained cells were SCs (Fig. 2), the isolated SCs (2.5×10^4 cells) were subcultured on collagen sheets (20 mm $\times$ 20 mm) (Atree Inc., Tokyo, Japan) in a longitudinal orientation in DMEM/F-12 and 2.5% FBS containing 1% Pen/Strep at 37°C in a humidified atmosphere of 5% CO_2. After a 2-week culture period, the oriented collagen sheets with the cultured SCs (Fig. 3) were rolled parallel to the longitudinal orientation and inserted longitudinally into a 20-mm collagen conduit (Atree), completing the fabrication of the oriented collagen scaffold containing cultured SCs.

Transplantation to the Deficient RLN

Experimental dogs were anesthetized as described above. A skin incision of approximately 10 cm was made in the midline of the neck, exposing the unilateral RLN in each canine. Twenty millimeters of each right RLN were resected roughly 10 mm distal to the lower border of the cricoid cartilage (Fig. 4A). Fabricated scaffold containing SCs was autotransplanted to each

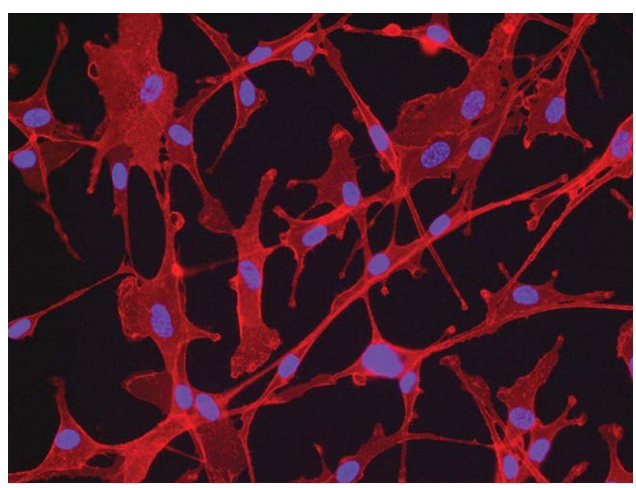

Fig. 2. Immunocytochemical findings of the cultivated cells. Smaller cells with high nucleus-cytoplasm ratios expressed S-100b protein.

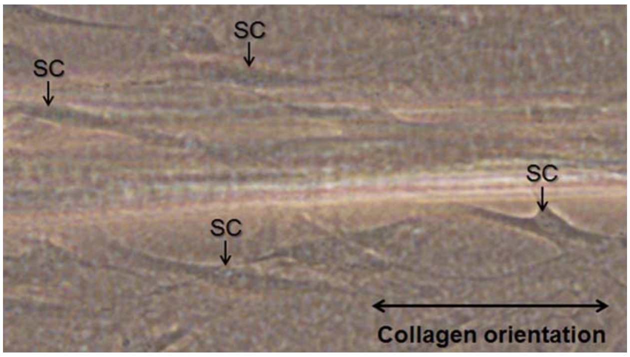

Fig. 3. Phase-contrast microscopy revealed Schwann cells in the same orientation as the collagen fibers on the collagen sheet. SC = Schwann cell.

deficient RLN between the proximal and distal nerve stumps. To improve fixation, 1 mm of both nerve stumps was inserted into both sides of the collagen conduit and fixed with absorbable surgical sutures and fibrin glue. Immobility of the unilateral vocal fold was confirmed endoscopically (Olympus, Tokyo, Japan) and the skin incision was closed.

Evaluation of the Transplanted RLN

After transplantation, endoscopy was performed every other week to check for spontaneous movement of the bilateral vocal folds under light general anesthesia without endotracheal intubation. Coordinated movement of the bilateral vocal folds was assessed according to the criterion for the clinical diagnosis of paresis[9,10] independently by two authors of this article. Atrophic changes of the vocal fold were assessed by the observation of the degree of bowing or concavity in the vocal folds. If spontaneous movement was absent in the half-awake state, stimulation of the arytenoid mucosa was attempted using the tip of the endoscope.

Six months after transplantation, bilateral RLNs, including those in the larynx and trachea, were removed. The RLNs were removed and fixed by 4% paraformaldehyde in 0.1 M PBS for 48 hours. The right RLN tissue was divided into three specimens; namely, the proximal stump, the distal stump, and the middle. These specimens were sectioned to a thickness of 4 μm and mounted on glass slides. Deparaffinized and hydrated sections were rinsed with PBS. Histological examinations were performed using hematoxylin and eosin (H&E) and elastic Van Gieson (EVG) stains and compared with the normal left RLN.

Immunohistochemical staining was also performed using deparaffinized and hydrated sections. Antigen retrieval by microwave irradiation was conducted using Target Retrieval Solution (S1699; Dako, Glostrup, Denmark) for 10 minutes. Sections were blocked with 3% hydrogen peroxide for 10 minutes. The specimens were then incubated at 4°C overnight with anti-myelin protein zero (P_0) (bs-0337R; BiossUSA, Woburn, MA) diluted 1:200. Myelin P_0, the major glycoprotein component of peripheral myelin, is normally expressed by mammalian SCs until myelination begins.[11] After rinsing with PBS and incubating with universal secondary antibodies conjugated with horseradish peroxidase-labeled amino acid polymer (Histofine Simple Stain MAX-PO; Nichirei, Tokyo, Japan) at room temperature for 30 minutes, immunoreactive proteins were stained with 3,3′-diaminobenzidine tetrahydrochloride for 5 to 10 minutes. Specimens were counterstained with hematoxylin, dehydrated with xylene, and mounted.

RESULTS

Immunocytochemical investigation revealed that cultured cells with a high nucleus-cytoplasm ratio expressed both S-100b protein (Fig. 2) and GFAP but not vimentin, and were identified as SCs. Phase-contrast microscopy revealed the orientation of SCs as consistent with that of the collagen fibers on the collagen sheet (Fig. 3). As mentioned above, the collagen sheet with the cultured SCs was rolled longitudinally and inserted into a 20-mm collagen conduit.

Following endoscopic confirmation in each canine that the right vocal fold was fixed in the intermediate position, the fabricated collagen conduit was successfully transplanted to the resected portion of the right RLN (Fig. 4B). One month after transplantation in all three canines, laryngeal endoscopy revealed no atrophy of the right vocal folds, despite remaining fixation of the right vocal fold in intermediate position in each of the dogs. Laryngeal stimuli to the mucosa via flexible endoscope did not evoke movement in the right vocal folds. Within 2 months after transplantation, coordinated adduction of

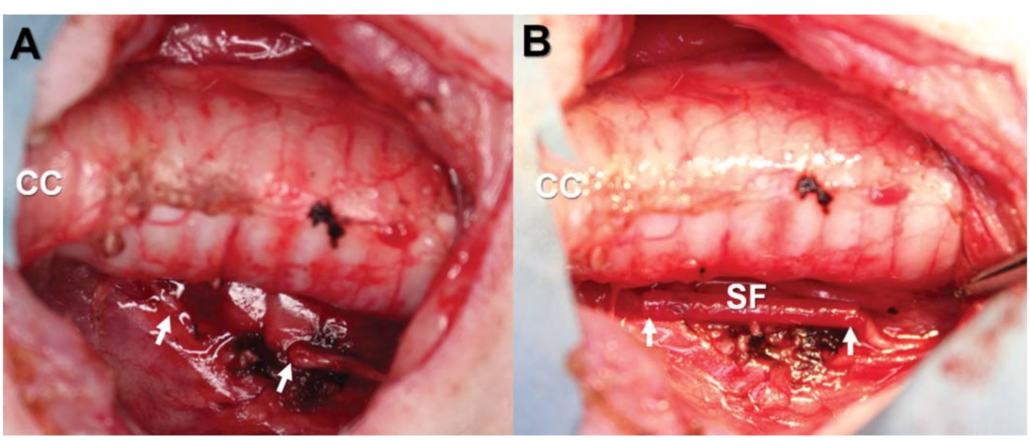

Fig. 4. Surgical findings. (A) Right recurrent laryngeal nerve resection of 20 mm. (B) Transplantation of the fabricated scaffold containing Schwann cells to the nerve defect. Arrows show nerve stumps. CC = cricoid cartilage; SF = a scaffold containing oriented Schwann cells.

Fig. 5. Fiberscopic findings of vocal fold movements. Coordinated adduction of the vocal folds with the vocal processes was identified only when stimuli were provided to the mucosa via the flexible endoscope.

the vocal folds with the vocal processes was observed only when stimulation was provided under light general anesthesia (Fig. 5). However, spontaneous motion of the right vocal folds was impaired and remained in the intermediate position under light general anesthesia. These phenomena continued 6 months after transplantation. Neither atrophy in the right vocal folds nor hoarseness was noted in any of the dogs.

Nerve regeneration was identified through the middle section of the transplanted tissue. H&E staining showed bundles of nerve fibers including axons (Fig. 6A). Several nerve fibers were surrounded by perineurium, whereas some were interspersed with collagen fibers. EVG staining showed no epineurium surrounding any nerve fibers. Regenerated axons of varying diameters were found, and these were thinner than the axons of the unresected side. We noted no neuroma formation or any other pathological reactions. Immunohistochemically, myelin P_0 was expressed in many myelin sheaths (Fig. 6B), indicating that the regenerated nerve was myelinated nerve fibers. Some cells spread around the nerve bundle were identified as isolated SCs.

DISCUSSION

The present study demonstrated that cultured SCs in the appropriate cell orientation can facilitate nerve regeneration along the axonal alignment, even after sectioning. To the best of our knowledge, we are the first to report partial functional regeneration of a 20-mm-long RLN defect in a canine model.

The RLN is known as a difficult peripheral nerve to regenerate functionally, even though experimental studies of RLN regeneration have reported successful morphological recovery. Several major reasons may explain why functional recovery has not been obtained. First, the RLN is long and thin, making it difficult for axonal regeneration to reach the original target organs. Second, the absence of nerve control to intrinsic laryngeal muscles for a long period can quickly lead to neurogenic atrophy of the muscles. Third, the simultaneous innervations of the four intrinsic laryngeal muscles with antagonistic functions, one as an abductor (the posterior cricoarytenoid muscle) and the other three as adductors (the lateral cricoarytenoid, the thyroarytenoid, and the transverse arytenoid muscles) can cause synkinesis after RLN neogenesis. To date, the preferred procedure choice for RLN reconstruction has been an autologous nerve graft, but even this has yielded unsatisfactory and disappointing results. The dense structure of the nerve graft may obstruct the growth of sprouting axons.

Recently, peripheral nerve regeneration using a variety of synthetic NGCs has been reported by several studies.[11–14] Most of these studies found that compared to autologous nerve grafts, artificial nerve guides produce a higher quality nerve regeneration. In particular,

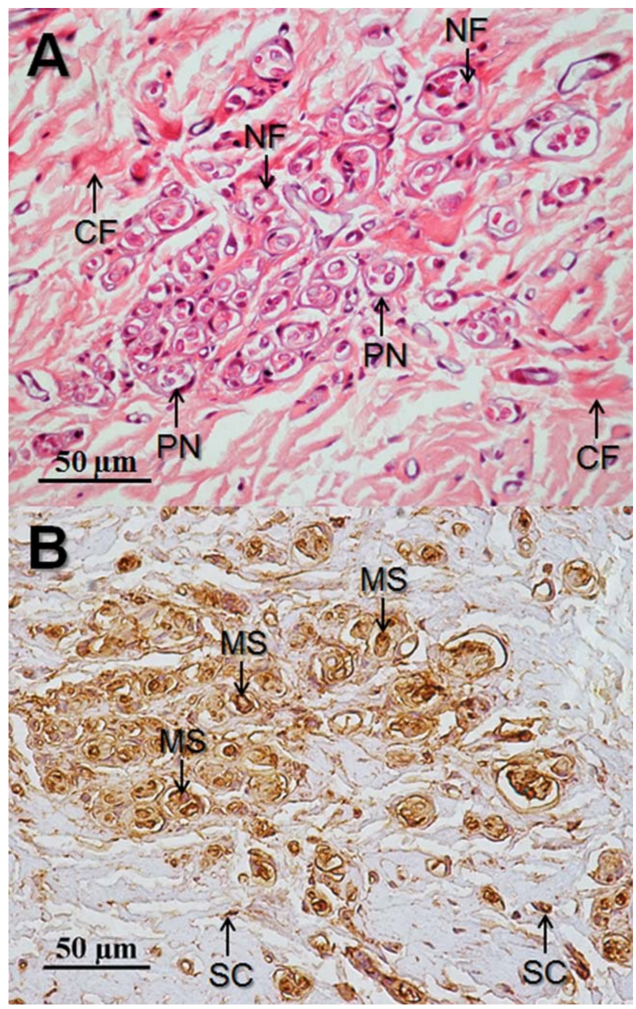

Fig. 6. Immunohistochemistry of the regenerated recurrent laryngeal nerve. (A) Hematoxylin and eosin. (B) Myelin protein zero. CF = collagen fiber; MS = myelin sheath; NF = nerve fiber; PN = perineurium; SC = Schwann cell.

collagen conduits[15,16] have been used to regenerate axons along the filaments, and some have used a polyglycolic acid (PGA) tube coated with collagen.[17,18] These artificial nerve guides are advantageous in their biodegradability, low antigenicity, and good affinity to the living tissue, and because they comprise an open structure through which regenerating nerve fibers can grow easily. Kanemaru et al. were the first to demonstrate that the RLN can be functionally regenerated using a PGA tube coated with collagen even after a 10-mm-long resection in canine models.[17] Furthermore, Nakamura et al. reported that the myelinated axons on the PGA-collagen tube were larger in diameter than those on the autograft side.[18] Therefore, biodegradable conduits coated with collagen could serve as effective alternatives to conventional autografting to repair RLN defects. However, regeneration of RLN defects longer than 10 mm using NGCs has yet to be reported in canine models.

Our preliminary study on RLN regeneration using SCs demonstrated that the transplantation of a scaffold including cultured SCs in the proper cell orientation successfully facilitated nerve regeneration along axonal alignment. SCs within the peripheral nerve system are essential for peripheral nerve repair following injury, and a lack of these cells can impede nerve regeneration.[19] Previously, other studies found that incorporation of SCs within NGCs can improve neuronal cell growth and orientation. However, injuries resulting in a nerve gap lack spatial guidance between the proximal and distal axons. At the basic research level, use of an artificial nerve guide containing viable SCs has been one of the most promising approaches for repairing nerve injuries. Moreover, providing a scaffold for SCs in the form of oriented biodegradable fibers can improve cell growth and subsequent axonal alignment.

To facilitate the development of these transplantation techniques using cultured SCs, a method to obtain large, essentially pure populations of SCs from autologous peripheral nerve must be developed. Unfortunately, SCs obtained from nerve biopsies are difficult to maintain and expand in a cell culture.[20] SCs differentiate into either myelinating or ensheathing cells in accordance with various signaling. The overgrowth of fibroblast-like cells should be restricted while culturing, and thus a reliable method for SC acquisition is necessary. Whereas a simple method to isolate mitotically active SCs in culture has been reported,[21,22] this previous method using mitogens is not clinically acceptable, and other methods that do not require mitogens have failed to isolate SCs effectively or require a long period of time. In contrast, predegeneration and subsequent culture was reported to both increase the number of SCs and restrict overgrowth of fibroblast-like cells.[23,24]

Our simplified method based on an established protocol with low FBS concentration[25] facilitated SC culturing, but required over a month to obtain large enough populations of SCs. Any therapeutic application would likely require that the cultures produce a sufficiently high number of SCs with a high level of purity in a relatively short period of time. An efficient system such as directed differentiation of bone marrow stromal cells[26,27] into derived SCs may be available for peripheral nerve regeneration in the future.

CONCLUSION

This experimental study successfully fabricated an oriented collagen scaffold containing cultured SCs, which resulted in the partial functional regeneration of a 20-mm-long RLN defect in a canine model. Within 2 months after the transplantation, coordinated adduction of the vocal folds with the vocal processes was observed only with external stimuli. In the context of functional RLN regeneration, cultured SCs in an appropriate cell orientation may facilitate axonal regrowth along the scaffold cell alignment even after RLN sectioning.

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